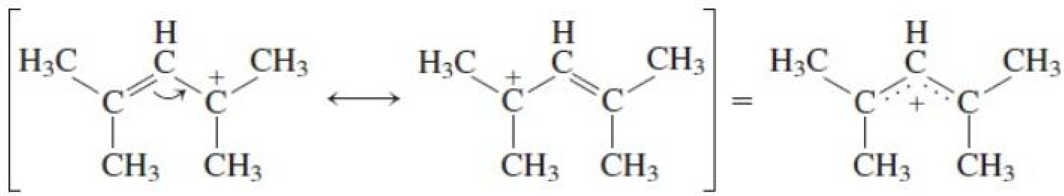
Solutions to Problems

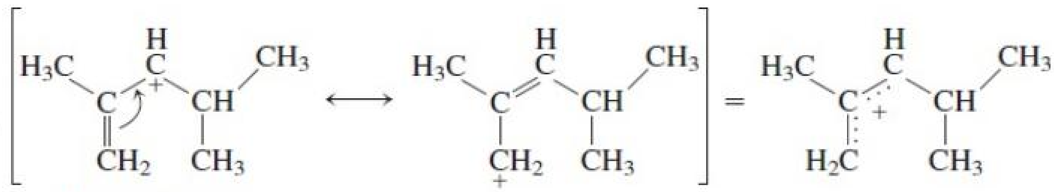
32 and **33.** Major contributing resonance forms are labeled.

(a)



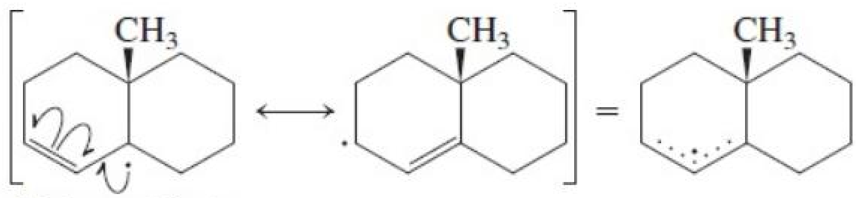
Equal contributors

(b)



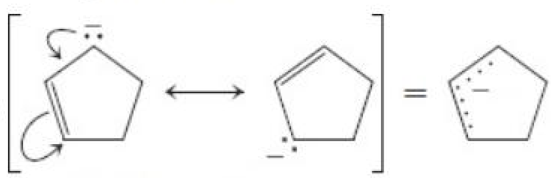
Major contributor
(charge on secondary carbon)

(c)



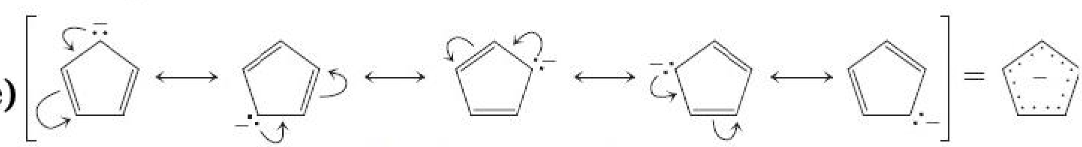
Major contributor
(tertiary radical-like)

(d)



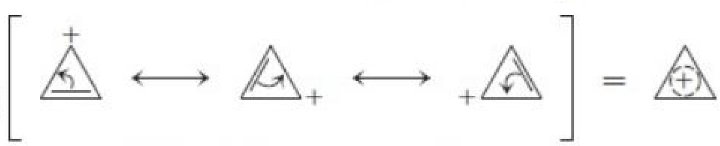
Equal contributors

(e)



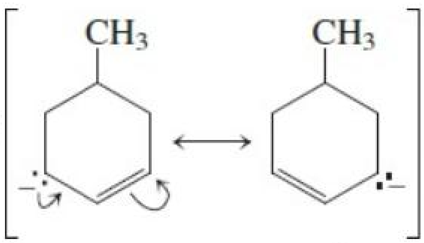
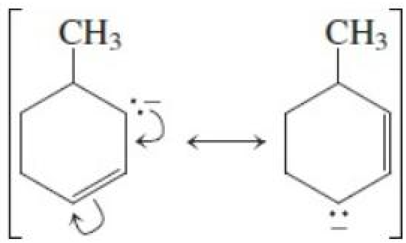
All contributors are equal

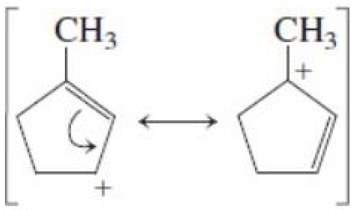
(f)



All contributors are equal

34. **(a)** [CH3C•HCH=CH2↔CH3CH=CHC•H2
$CH_3\dot{C}HCH$â $CH_2 \leftrightarrow CH_3CH$â $CH\dot{C}H_2$]

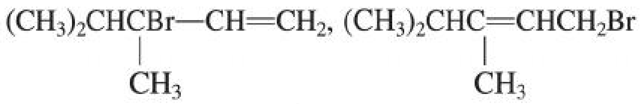
(b)  or 

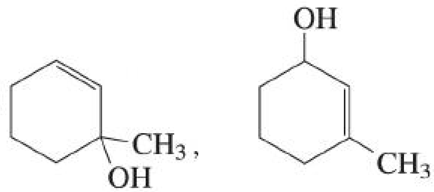
(c) 

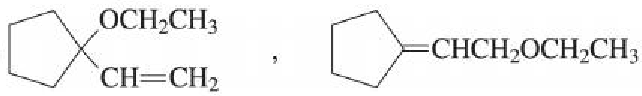
35. Radicals: allylic > tertiary > secondary > primary
Radicals : allylic > tertiary > secondary > primary

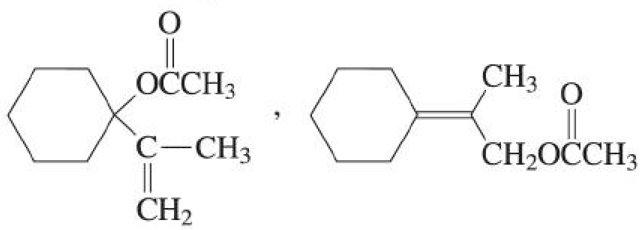
Cations: tertiary > allylic ≈ secondary > primary
Cations : tertiary > allylic ≈ secondary > primary

Hyperconjugation, which is at least partially responsible for the tertiary > secondary > primarytertiary > secondary > primary stabilization order, is more important for cations than for radicals. The effect is large enough for tertiary cations to exceed resonance-stabilized allylic cations in stability (the reverse of the order for radicals).

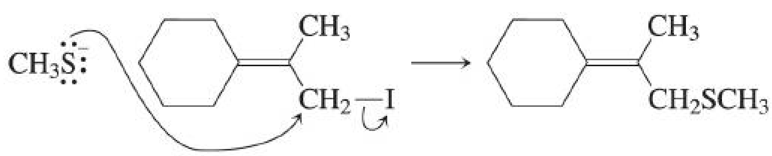
36. **(a)** 

(b) 

(c) 

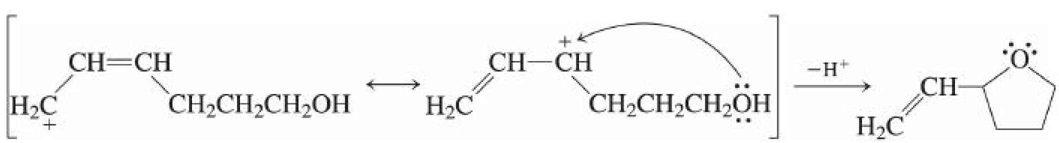
(d) 

(e) Different! SN2,S_N2, not SN1S_N1 conditions:



This is the only product.

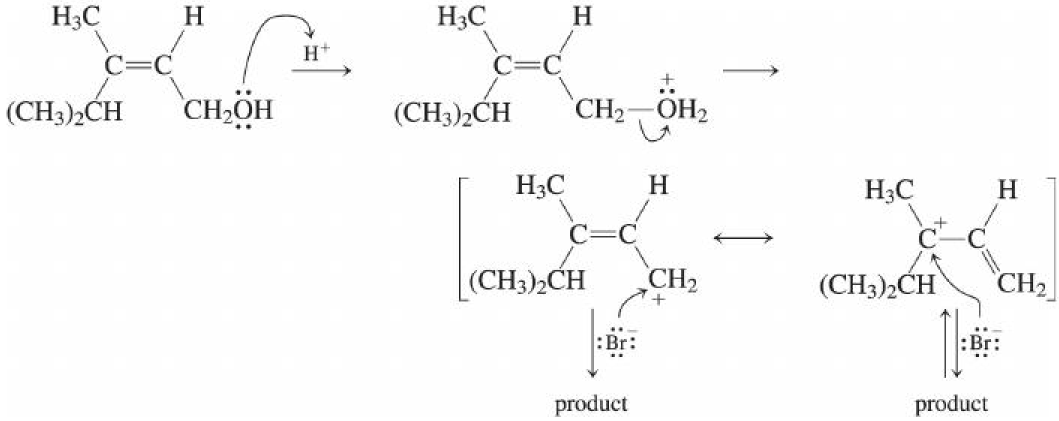
(f) Intramolecular version:



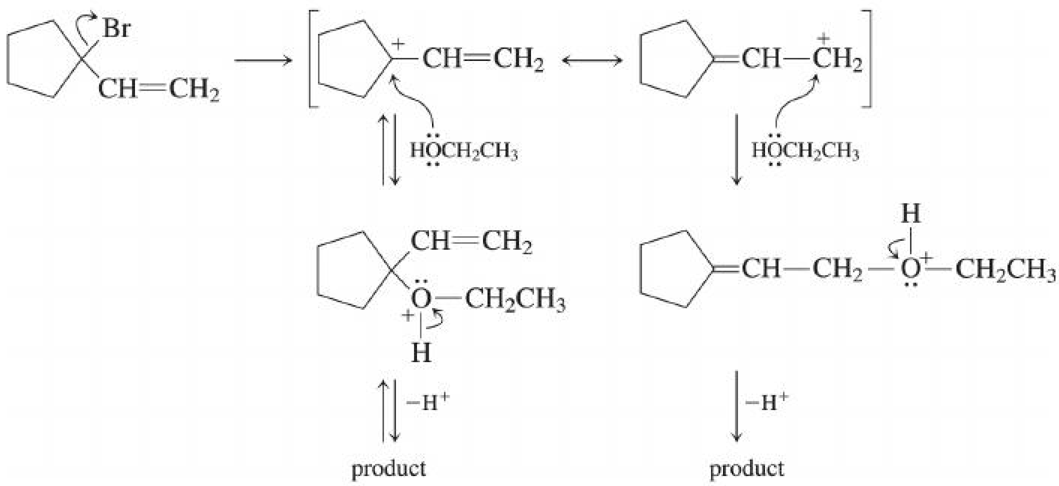
Again, only one product forms. Of the two resonance forms of the allylic cation, the second, with the positive charge on a secondary carbon, is the more important contributor. Attack by the oxygen nucleophile at this position is therefore kinetically favored. In addition, ring closure to give a five-membered ring is thermodynamically preferable to bond formation at the other end, which would give a more strained seven-membered ring.

37.

(a)



(c)



(e), (f) See answers to Problem 36.

38. **(a)** Tertiary > allylic ≈ secondary ≫ primary (order of cation stability)
Tertiary $>$ allylic $\approx$ secondary $\gg$ primary (order of cation stability)

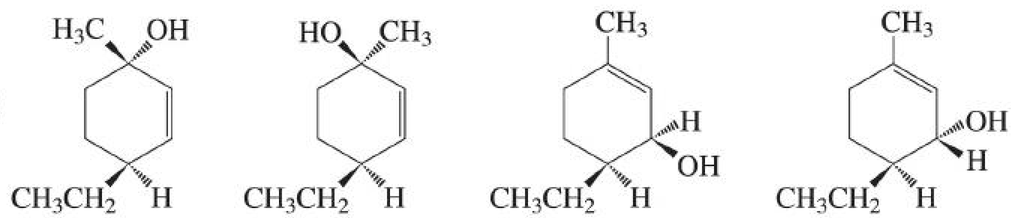
(b) Allylic > primary > secondary ≫ tertiaryAllylic $>$ primary $>$ secondary $\gg$ tertiary

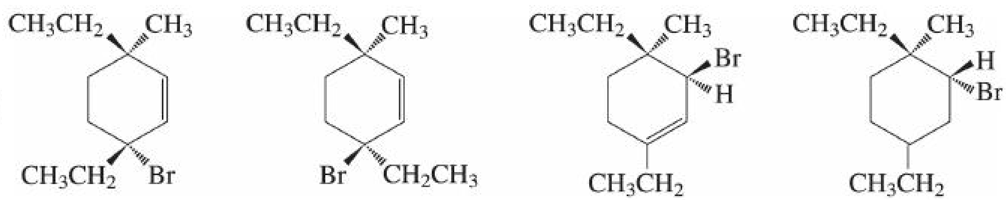
39. SN1S_N1 reactivity: e (allylic and tertiary) > a$>$ a (allylic and secondary) > d$>$ d (forms same cation as e, but requires ionization at primary carbon, so will be slower) > c> b> f$>$ c $>$ b $>$ f (these follow cation stability order) SN2S_N2 reactivity: Steric hindrance predominates, so f > b> d> c> a> ef $>$ b $>$ d $>$ c $>$ a $>$ e

40. SN2S_N2 reactivities: Data in this chapter (Section 14-3) reveal that allylic halides are about 102$10^2$ more reactive than their non-allylic counterparts in SN2S_N2 displacements. Therefore, all the primary allylic systems (b, c, d, and f) will be more reactive than a saturated primary halide—even the branched system (c) will possess higher reactivity, because branching reduces reactivity only by about a factor of 20 (Table 6-9). The secondary allylic system (a) will be similar in reactivity but perhaps a bit slower than a saturated primary—secondary systems are more than 102$10^2$ slower than primaries (Table 6-8), so the steric hindrance of the greater substitution just about cancels the acceleration due to the allylic system. Both allylic and saturated tertiaries are quite dead to SN2S_N2 displacement.

SN1S_N1 reactivities: Follow cation stabilities together with nature of the position of the leaving group. So (e) is fastest, followed by a saturated tertiary halide. Then comes (a), followed by the primary allylic systems (probably in the order d, c, b, and f), which are comparable to the saturated secondary. Saturated primary halides do not react by the SN1S_N1 mechanism.

41. Write all possible allylic isomers in each case and pay attention to stereochemistry.

(a) 

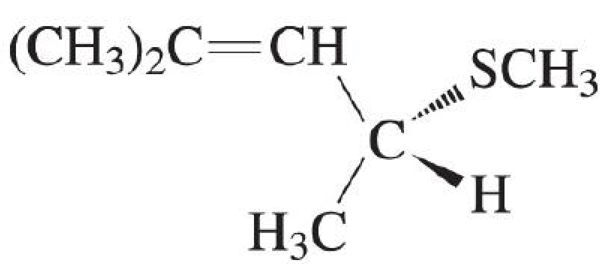
(b) 

(c) $CH_3CH_2C(CH_3)(Br){-}CH{=}CH_2$ (racemic) $\quad CH_3CH_2C(CH_3){=}CHCH_2Br$

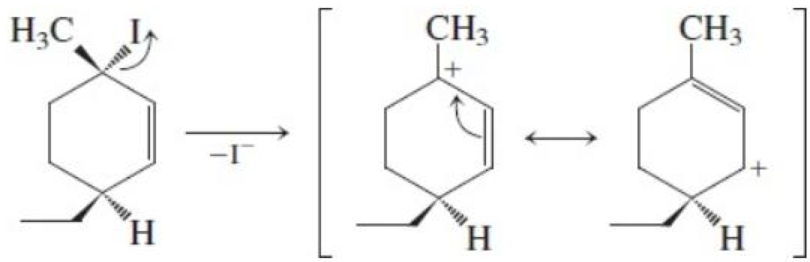
(d) [CH3C⁻HCH=CHCH2CH3↔CH3CH=CHC⁻HCH2CH

$[CH_3\overline{C}HCH = CHCH_2CH_3 \leftrightarrow CH_3CH = CH\overline{C}HCH_2CH_3]Li^+$

(e) $CH_3CH(CH_3CHOH)CH{=}CHCH_2CH_3 + CH_3CH{=}CHCH(CH_3CHOH)CH_2CH_3$ (All possible stereoisomers for each structure)

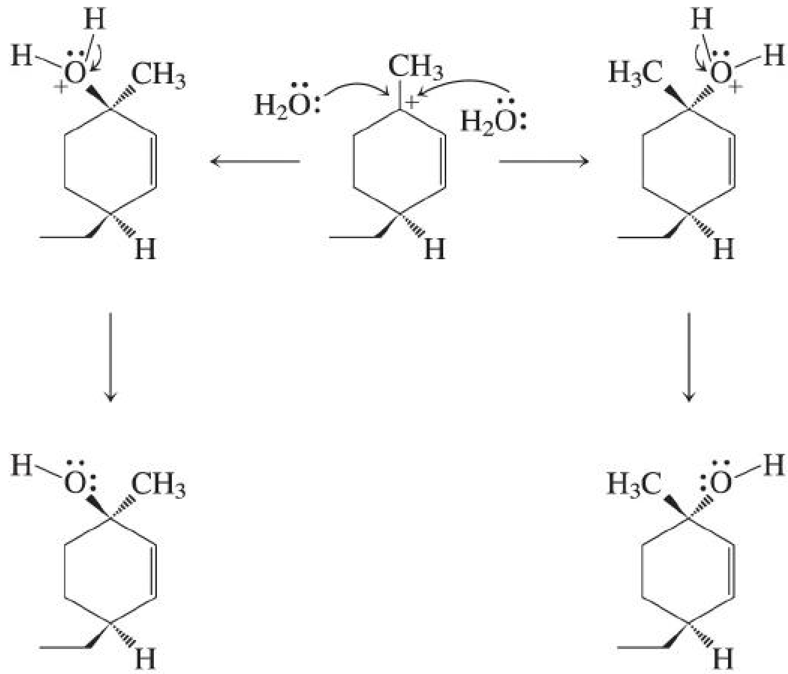
(f) 

42. The substrate is a tertiary, allylic halide, and the leaving group (iodide) is a good one. The solvent is water, an excellent one for formation of ions, suggesting that we should begin the process in the manner of an SN1S_N1 substitution: departure of the leaving group. The result is a carbocation which, being allylic, has two resonance forms:

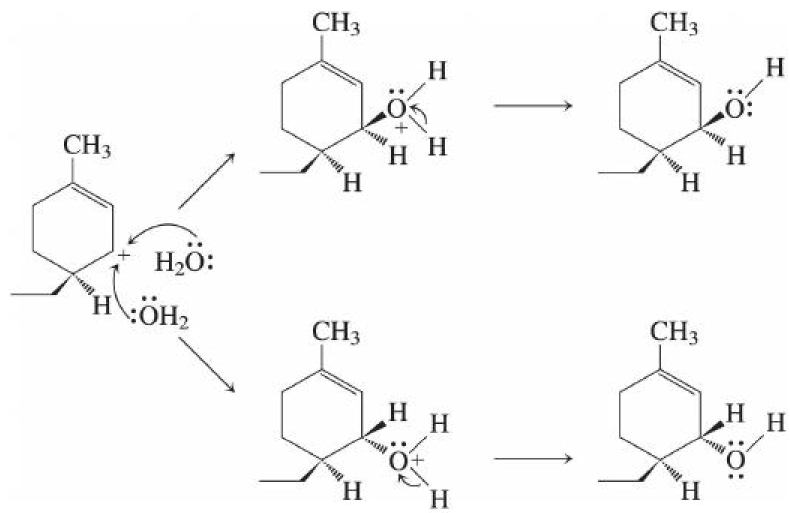


When we proceed to the next step, we will therefore have two possible positively charged positions to which the nucleophile, water, can add.

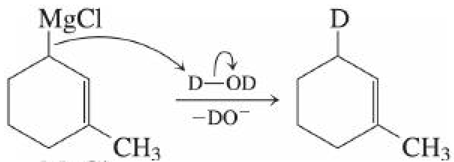
Notice that the carbon that originally contained the halogen is now trigonal planar. **Thus we no longer depict the methyl group with a wedged bond**; it has moved into the same plane as the other two bonds to this trigonal carbon atom. This carbocation still contains one stereocenter, however: the carbon with the ethyl substituent. Because of this stereocenter, the two faces of the cyclohexane ring are stereochemically distinct. The addition of water can take place to the same face or the opposite face of the one containing the ethyl group. For clarity, we first draw addition of water to the carbon that originally contained the leaving group (followed by proton loss to give the final products):



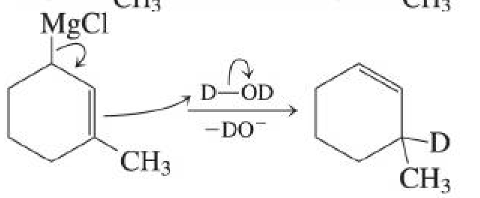
Water can similarly add to the other end of the allylic carbon, the other position with positive charge, and again to either face of the ring, relative to the ethyl group:



Thus, in all, four isomeric products are possible.

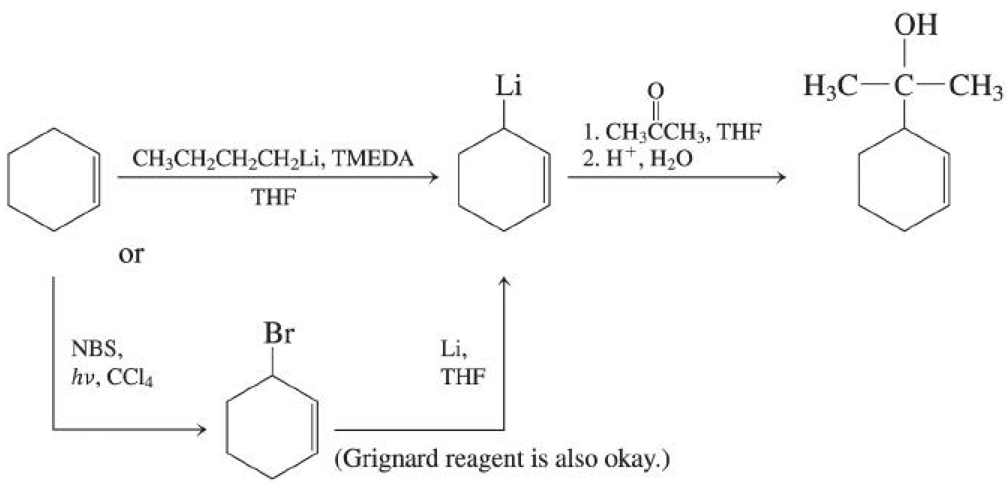
13. 

and



Grignard reagents behave like carbanions. Being an allylic Grignard reagent, this species will behave as a nucleophile and (here) a base at both ends of the allylic system.

14.



15. **(a)** cis-2-trans-5-Heptadiene, or (2E, 5Z)-2,5-heptadiene
cis-2-*trans*-5-Heptadiene, *or* (2*E*, 5*Z*)-2, 5-heptadiene

(b) 2,4-Pentadien-1-ol2, 4-Pentadien-1-ol

(c) (5S, 6S)-5,6-Dibromo-1,3-cyclooctadiene
(5*S*, 6*S*)-5, 6-Dibromo-1, 3-cyclooctadiene

(d) 4-Ethenylcyclohexene4-Ethenylcyclohexene

16. $H_2C{=}CH{-}\overset{\overset{\displaystyle H}{|}}{C}H{-}CH{=}CH_2$ 1,4-Pentadiene has the weakest C–HC – H
bond (arrow), a bond that is **doubly** allylic (DH◦~77 kcal mol−1);

$(DH^\circ \sim 77\,\mathrm{kcal\,mol^{-1}})$; this isomer will therefore be brominated fastest. Because only a very weak C–H$\mathrm{C-H}$ bond needs to be broken, its first propagation step has a much smaller EaE_a relative to 1,3-pentadiene, where a stronger methyl C–H$\mathrm{C-H}$ bond needs to be broken. However, both will give identical product mixtures because **identical** radicals are formed from each:

$$\mathrm{H_2C{-}CH{-}\dot{C}H{-}CH{-}CH_2} \equiv [\mathrm{H_2\dot{C}{-}CH{=}CH{-}CH{=}CH_2} \longleftrightarrow$$
$$\mathrm{H_2C{=}CH{-}\dot{C}H{-}CH{=}CH_2} \longleftrightarrow \mathrm{H_2C{=}CH{-}CH{=}CH{-}\dot{C}H_2}]$$

47. We figured we'd ask you this question now, so you could take your time and figure out the right answer instead of maybe getting it wrong on an exam.

Have a look at Figure 14-8 in the text. At high temperature, an equilibrium mixture exists because there is enough energy for molecules to "move" from any location on the reaction coordinate to any other location on it. In other words, all three species—the two products and the intermediate allylic cation—are interchanging rapidly, and at any given time the relative quantities of each are governed by their relative thermodynamic stabilities.

That being the case, if the temperature were to drop, the interconversion processes would slow down because fewer molecules would contain sufficient energy to pass over the activation barriers. This would mainly affect conversion of the two product molecules into the intermediate carbocation because those processes possess the highest activation barriers. The result is that the thermodynamic ratio of products originally established at high temperature would remain pretty much unchanged (frozen) upon cooling of the reaction mixture. It will **not** revert to the kinetic ratio!

18.

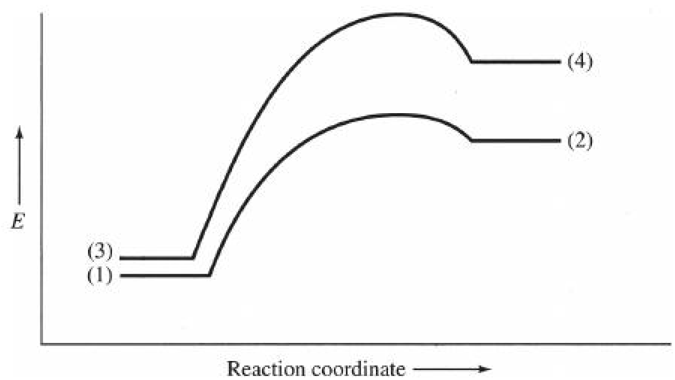
$$H_2C{=}CH{-}CH{=}CH{-}CH_3 \xrightarrow{H^+} H_3C{-}\overset{+}{C}H{-}CH{=}CH{-}CH_3$$

(1) Conjugated diene — (2) Allylic cation, secondary at each end

$$H_2C{=}CH{-}CH_2{-}CH{=}CH_2 \xrightarrow{H^+} H_2C{=}CH{-}CH_2{-}\overset{+}{C}H{-}CH_3$$

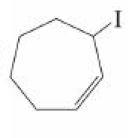
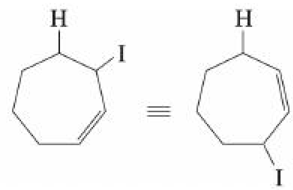
(3) Isolated diene — (4) Ordinary secondary cation

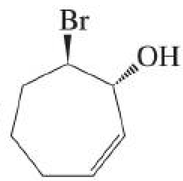
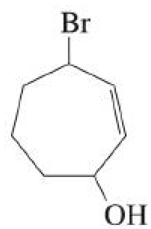
(1) is more stable than (3), and (2) is more stable than (4).

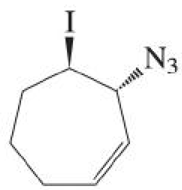
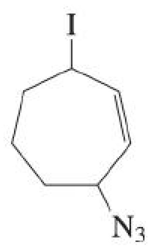


Reaction (1)+H+→(2)$(1) + H^+ \rightarrow (2)$ is faster and leads to the more stable cation. Note: When the text says that allylic and secondary cations are similar in energy, it is referring to the ease of formation of the simplest allylic cation, H2C+−CH=CH2,$H_2\overset{+}{C} - CH = CH_2$, which is primary at each end. Additional alkyl groups on allylic cations increase their stability and their ease of formation, as you might expect.

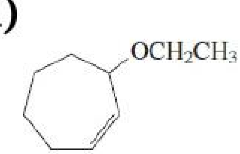
19. Expect 1,2- and 1,4-addition to occur in each case. Note that the 1,2-additions in (b) and (c) might be expected to show *anti* stereochemistry, similar to additions to ordinary alkenes.

(a)  1, 2 and 1, 4 products are the same!  (Both of these are 3-iodo-1-cycloheptene)

(b)  and 

(c)  and 

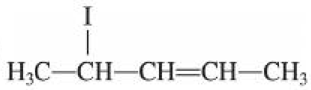
(d)

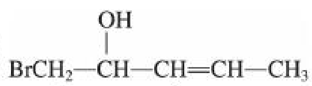
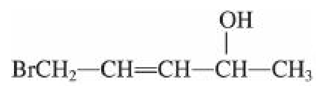


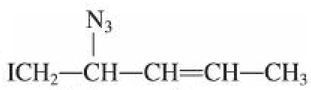
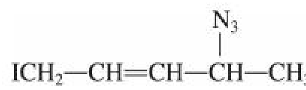
From both 1, 2- and 1, 4-additions

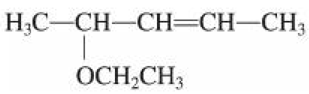
Note in parts (b) and (c) that, while 1,4-addition to conjugated dienes is stereorandom, 1,2-addition is often stereospecific, suggesting a halonium ion-like intermediate. See structures B and C in Exercise 14-13.

50. Addition of the electrophile will always be at C1, C1, generating the best allylic cation. The 1,2-addition product is given first.

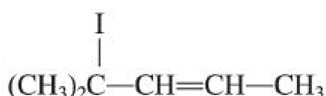
(a)  (cis and trans)

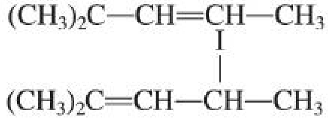
(b)  and  (cis and trans)

(c)  and  (cis and trans)

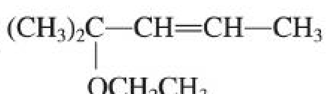
(d)  (cis and trans)

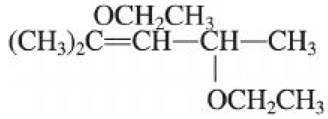
51.

(a)  (cis and trans) and

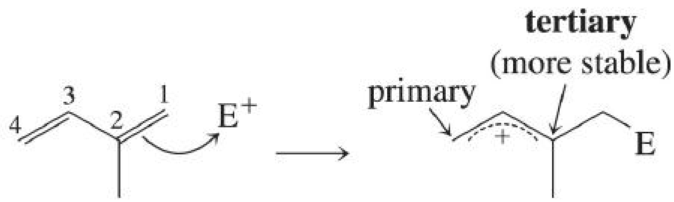


(b), (c) Same answers as Problem 50, but with a methyl group added to C2C2 in each case.

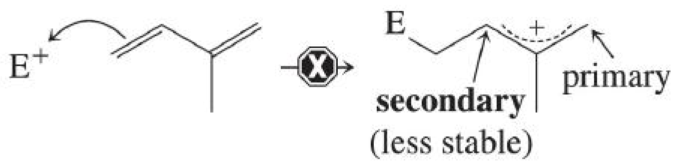
(d)  (cis and trans) and



52. In each case the electrophile adds to give the more stable allylic cation. Thus, addition to C1C1 is observed:

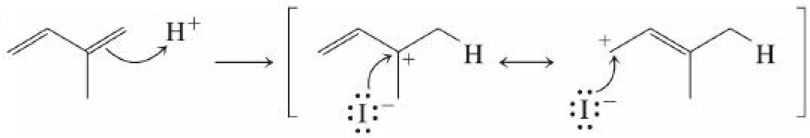


in preference to addition to C4:C4:

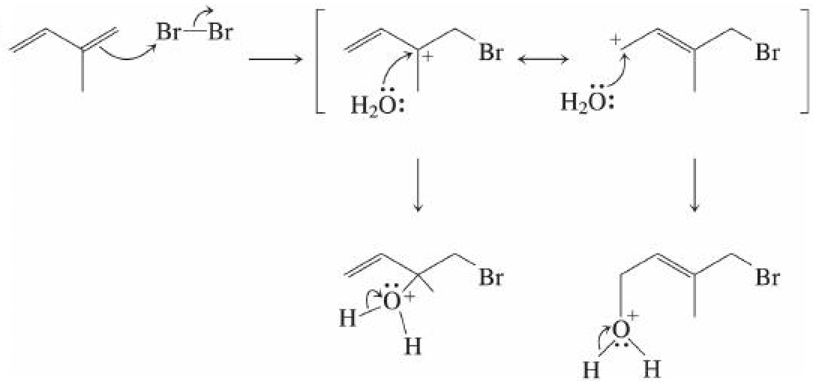


Final products in each case have already been given in the solution to Problem 51.

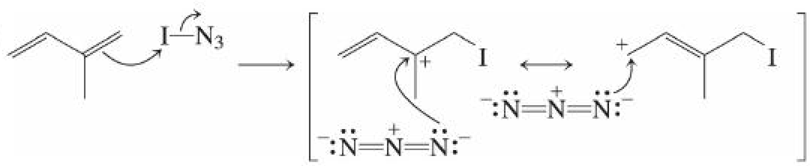
(a)



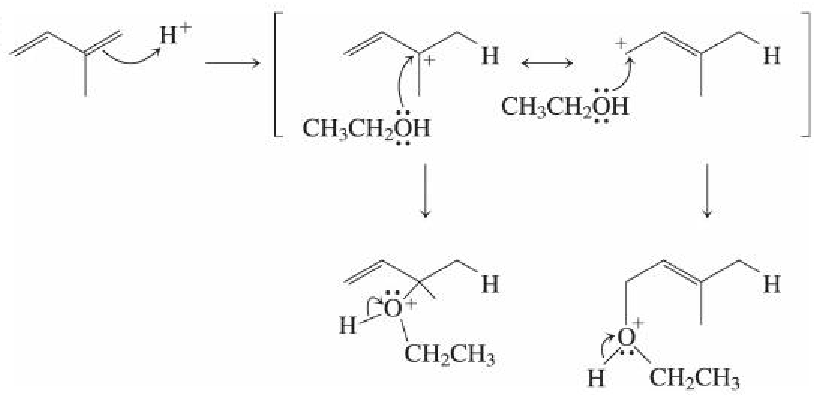
(b)



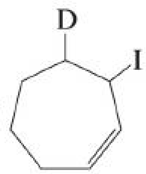
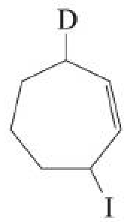
(c)

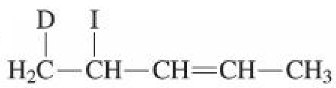
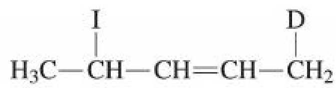


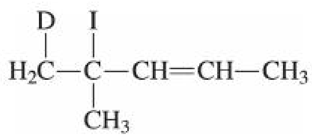
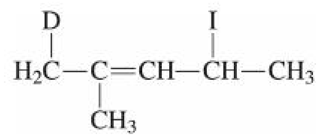
(d)



53.

(a)  and 

(b)  

(c)  and 

With DIDI it is easy to distinguish between 1,2- and 1,4-addition in the case of the cyclic diene in Problem 49 and the unbranched acyclic diene in Problem 50.

54. **(e)** [H2C═CH—C+H—CH═CH2↔H2C+—CH═CH—CH═CH2↔H2C═CH—CH═CH—C+H2] >

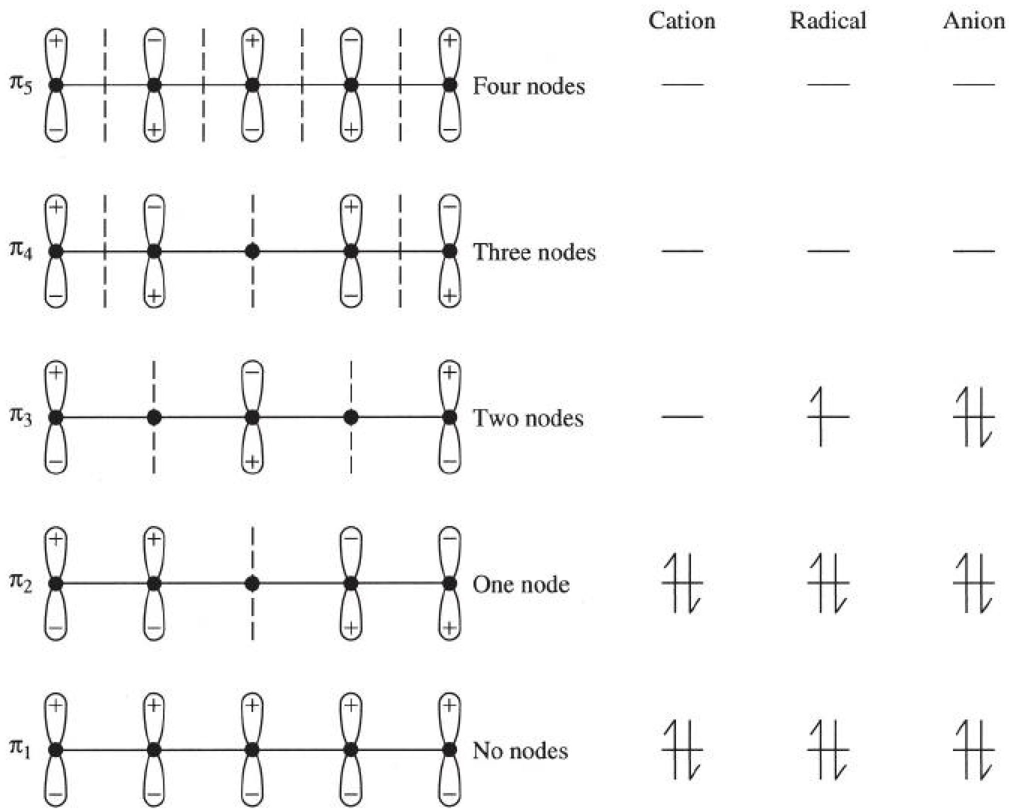
[H_2Câ CH—$\overset{+}{C}$H—CHâ CH_2 ↔ $H_2\overset{+}{C}$—CHâ CH—CHâ CH_2 ↔

H_2Câ CH—CHâ CH—$\overset{+}{C}H_2$] > **(d)** [H3C—C+H—CH═CH—CH3↔H3C—CH═CH—C+H—CH3] (secondary allylic at both ends)>

[H_3C—$\overset{+}{C}$H—CHâ CH—CH_3 ↔ H_3C—CHâ CH—$\overset{+}{C}$H—CH_3] (secondary allylic at both ends) >

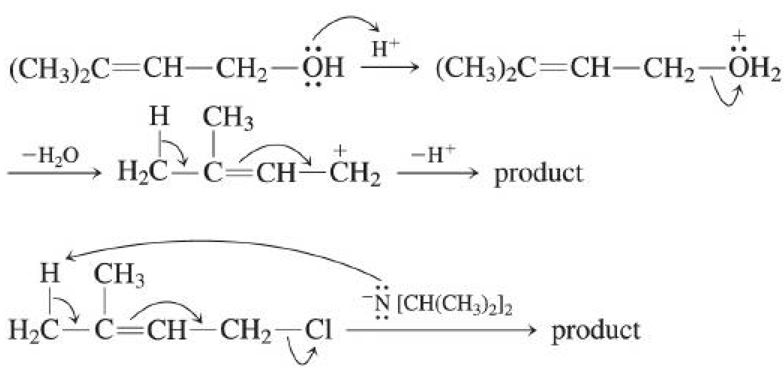
(a) H2C+—CH═CH2>$H_2\overset{+}{C}$—CHâ CH_2 > **(c) > (b)**(c) > (b)

55.

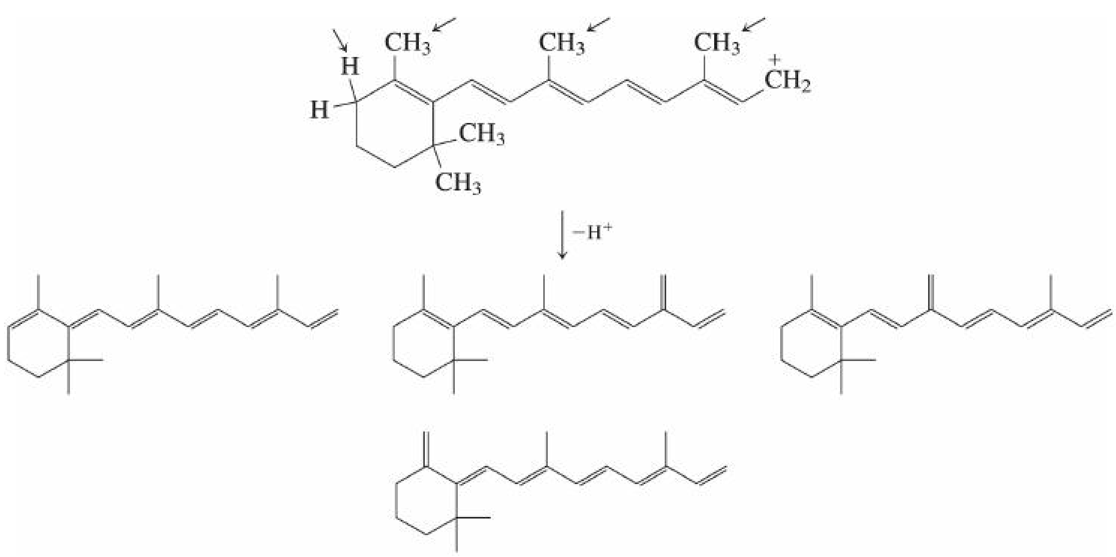


See the answer to problem 54(e) for the resonance forms of the cation and the answer to Problem 46 for the resonance forms of the radical.

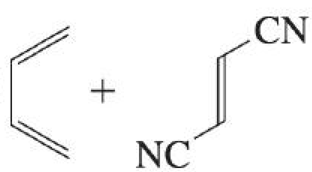
56.

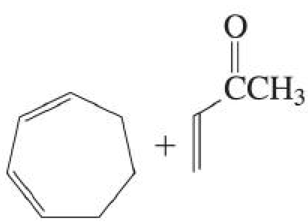


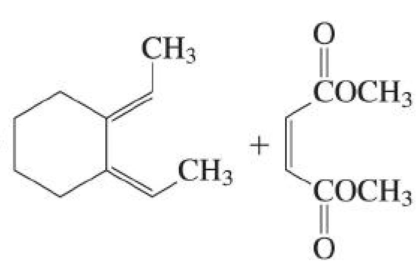
57. Any allylic hydrogen may be lost from the intermediate cation (arrows):

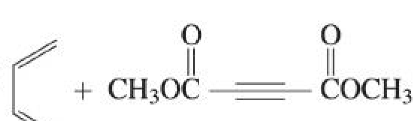

CH3
CH3
CH3
H
H
CH2
CH3
CH3
−H+

58.

(a) 
CN
NC

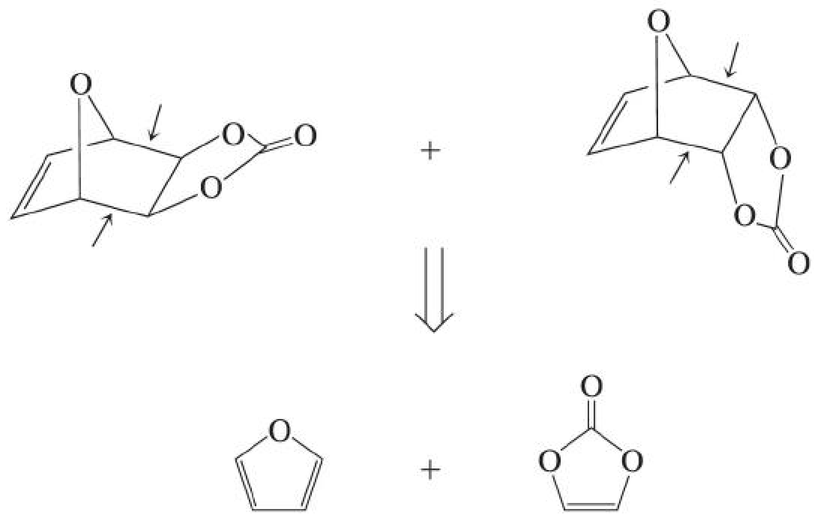
(b) 
O
CCH3

(c) 
CH3
CH3
O
COCH3
COCH3
O

(d) 
O
O
CH3OC
COCH3

59.

(a) Identify the bonds in the products formed in the Diels-Alder reaction. Then work backward to the starting molecules.



diene dienophile

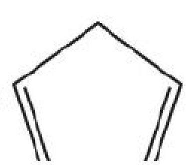
(b) See above.

(c) B is the *endo* adduct and is favored, as is expected for the typical Diels-Alder process.

60.

(a) H2C═CH—CH2—OCH3$H_2Câ\,CH—CH_2—OCH_3$

(b) H2C═CH—CHBr—CH3+CH3—CH═CH—CH2Br (cis trans)$H_2Câ\,CH—CHBr—CH_3 + CH_3—CHâ\,CH—CH_2Br$ (cis and trans)

(c) 

(d)

CH3CH2CHClCH═CHCH3+CH3CH2CH═CHCHClC
as mixtures of stereoisomers)

$CH_3CH_2CHClCH$â $CHCH_3 + CH_3CH_2CH$â $CHCHClCH_3$

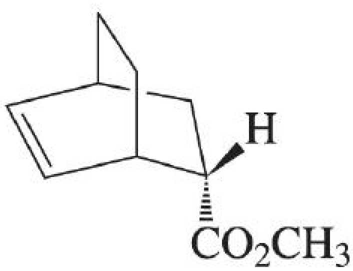
(both as mixtures of stereoisomers)

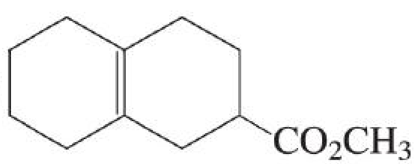
(e)

CH3CHBrCHOHCH═CHCH3+CH3CHBrCH═CHCH
as mixtures of stereoisomers)

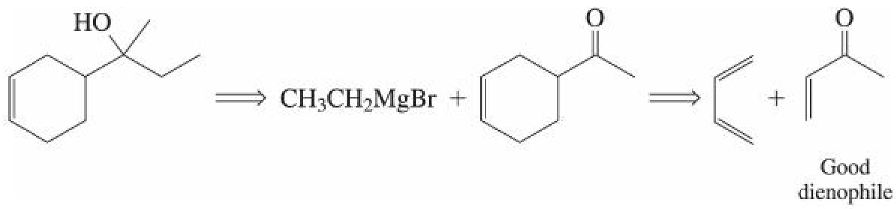
$CH_3CHBrCHOHCH$â $CHCH_3 + CH_3CHBrCH$â $CHCHOHCH_3$

(both as mixtures of stereoisomers)

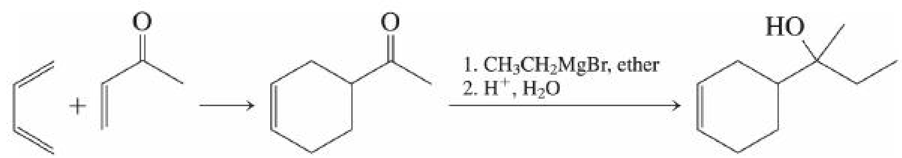
(f) 

(g) 

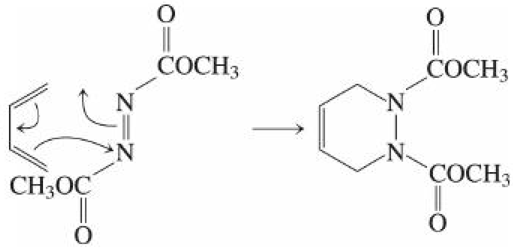
1. Taking note of the hint, recall from Section 14-8 in particular that alkenes containing electron-withdrawing groups are especially good dienophiles in Diels-Alder reactions. So,

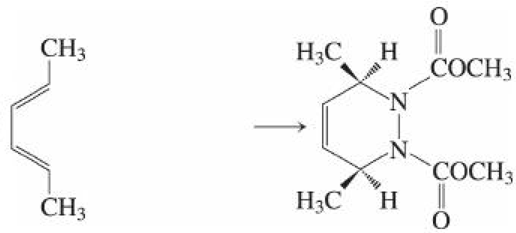


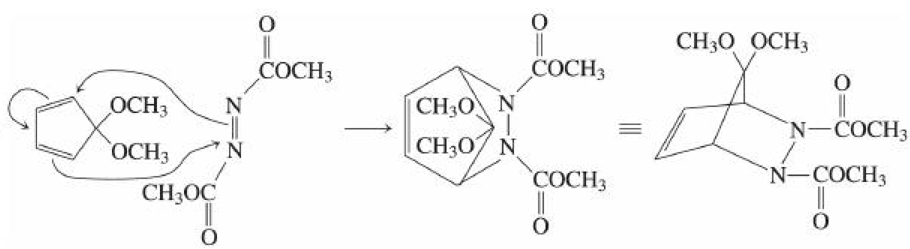
The synthesis, therefore, would go as follows:

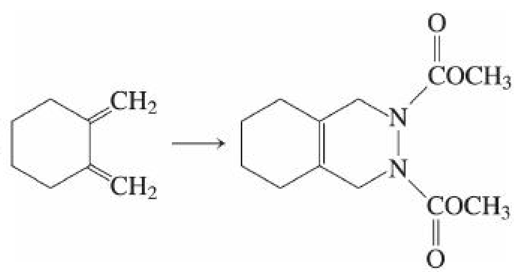


52.

(a) 

(b) 

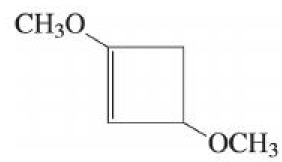
(c) 

(d) 

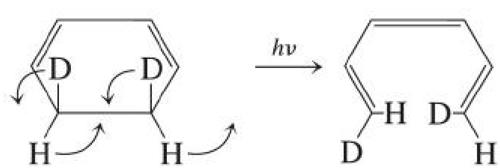
53. Diene A is similar in structure to 1,3-cyclohexadiene, which reacts well in Diels-Alder cycloadditions (Table 14-1). In diene B, however, the ends of the diene are locked in a zigzag conformation (called "s-trans") that puts the end carbons too far apart to bond to the two alkene carbons of a dienophile. Figure 14-9 illustrates how the Diels-Alder reaction involves a diene in a **U**-shaped conformation (called "s-cis"), which puts its end carbons close together. Dienes (like B) that cannot achieve this conformation will not participate with dienophiles in Diels-Adler reactions.

54. All are electrocyclic reactions. If you are a fan of American football and know that a **T**ouch**D**own is worth **6** points, then you are eligible to learn a simple mnemonic, namely, that a **6**-electron **T**hermal electrocyclic reaction is **D**isrotatory. Changing either from thermal to photochemical or the number of electrons by ±2±2 changes the rotation direction.

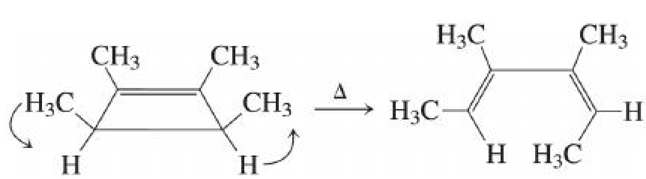
(a) A photochemical 1,3-diene ring closure (4 electrons). [Mechanism is disrotatory (Figure 14-11), although this particular diene does not contain substitution at both ends, which is necessary for the disrotatory outcome to be directly observed.]



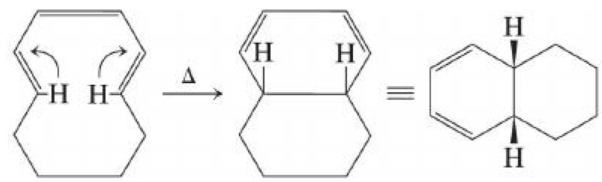
(b) Photochemical cyclohexadiene (6 electrons) ring opening, which is **conrotatory.**



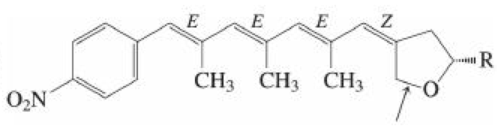
(c) Thermal cyclobutene (4 electrons) ring opening, which is **conrotatory.**



(d) Thermal hexatriene (6 electrons) ring-closure: **disrotatory.**



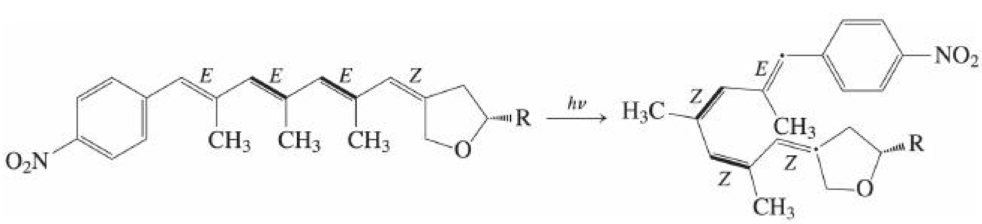
55.

(a) 

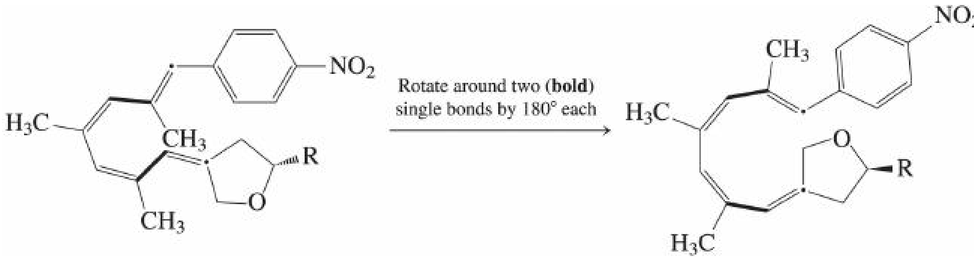
(b) The compound is highly conjugated: It is a tetraene in conjugation with a benzene ring. The two tetraene structures in Table 14.3 have λmax$\lambda_{\max}$ in the 300–400 nm$300–400\ \mathrm{nm}$ region, so it is not surprising that this compound does as well.

(c) The bond indicated by the arrow is allylic and will be most easily cleaved.

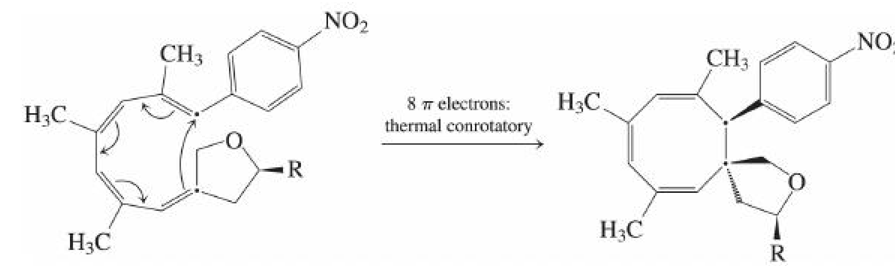
(d) This part is hard, because getting the molecule into the right shape for the subsequent reaction is tricky. You have to break it down into individual steps as follows. Let's hold the ether ring still at the right. Then isomerize the two (**bold**) alkenes from EE to ZZ. This is what you get:



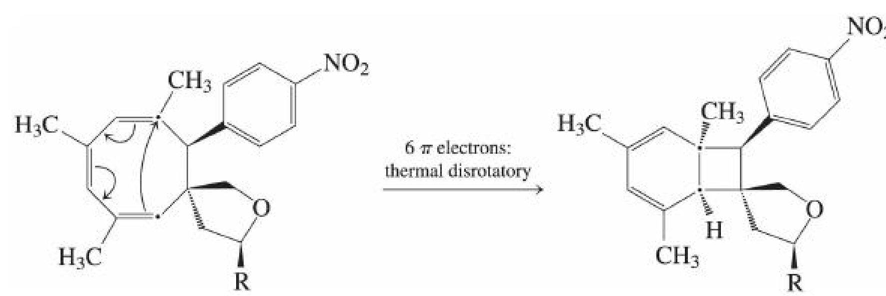
The tetraene is not quite arranged for ring-closure: The ends (dots) are pointing away from each other. We need to rotate around two single bonds (**bold**) first so that they are close:



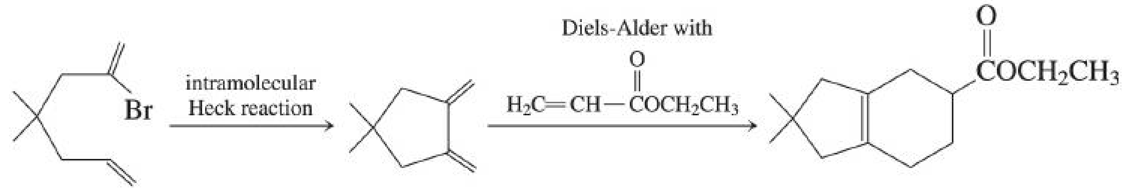
I've distorted the bond angles in the picture so that you can see everything. Actually the dotted carbons are pretty close. Now we are ready to do the first 8-π-electron8-π-electron reaction. By our rules, it is conrotatory. We make an eight-membered ring with three double bonds:

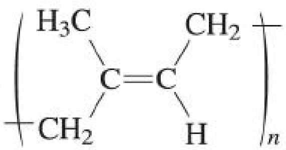


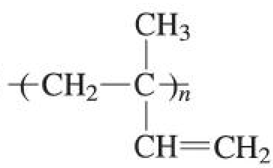
With the eight-membered ring closed, we're ready for the next step, a 6-π-electron6-π-electron ring closure involving the three double bonds. Again, the atoms that connect are indicated by dots. Now we have a disrotatory process, making a cyclohexadiene:

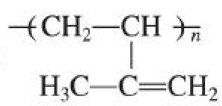


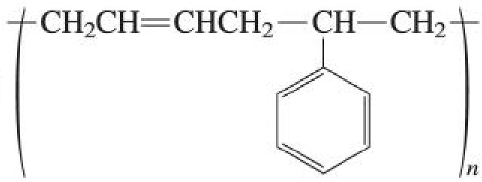
The product goes by the label SNF4435CSNF4435C and possesses both immunosuppressive and anti-cancer activity.

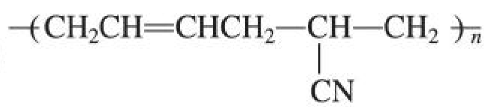
56. 

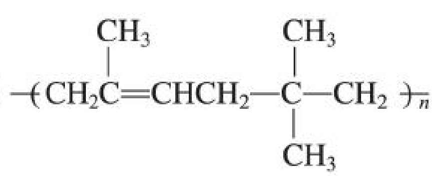
57. (a) 

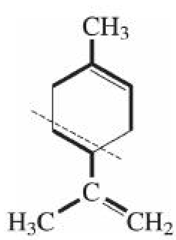
(b) 

(c) 

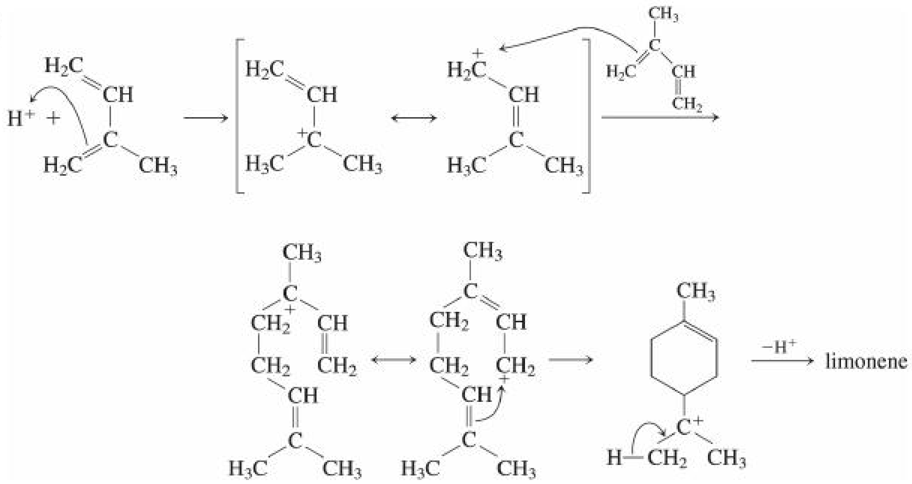
(d) 

(e) 

(f) 

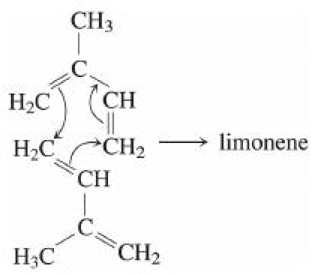
58. 

(a)



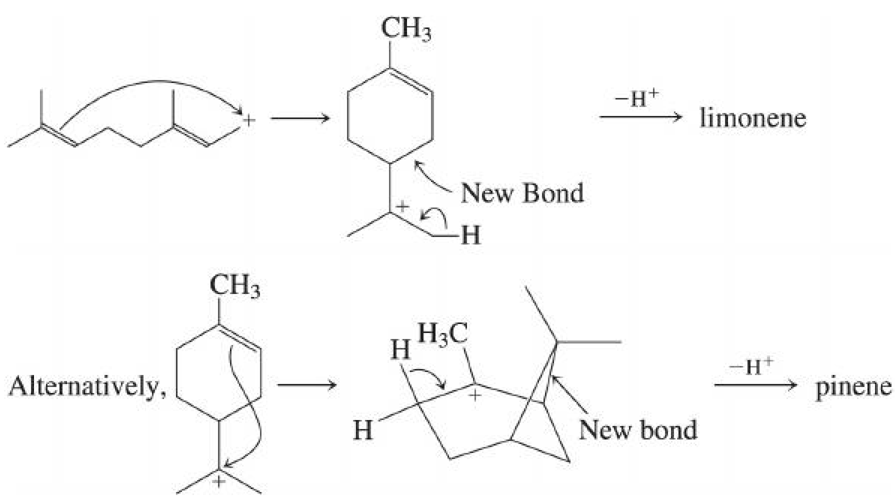
Notice that both additions occur to give an allylic cation that is tertiary at one end.

(b)



Via a Diels-Alder concerted cycloaddition reaction

69.



70. In each case the process involves promotion of an electron in the highest occupied molecular orbital (n or π)$(n \text{ or } \pi)$ to the lowest unoccupied molecular orbital (e.g., π*).$(\text{e. g. }, \pi^*)$. Use subscripts when there is more than one orbital of a given type.

(a) π→n (alternatively, π1→π2);
$\pi \rightarrow n$ (alternatively, $\pi_1 \rightarrow \pi_2$);

(b) π→n* (alternatively, π2→π3);
$\pi \rightarrow n^*$ (alternatively, $\pi_2 \rightarrow \pi_3$);

(c) n→π*;$n \rightarrow \pi^*$;

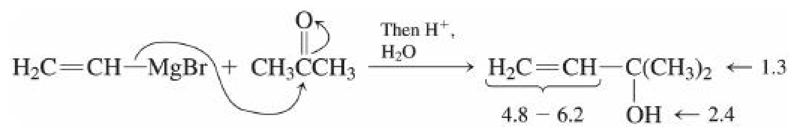
(d) n→π* (more specifically, n→π3*);
$n \rightarrow \pi^*$ (more specifically, $n \rightarrow \pi_3{}^*$);

(e) n→π* (alternatively, π3→π4*);
$n \rightarrow \pi^*$ (alternatively, $\pi_3 \rightarrow \pi_4{}^*$);

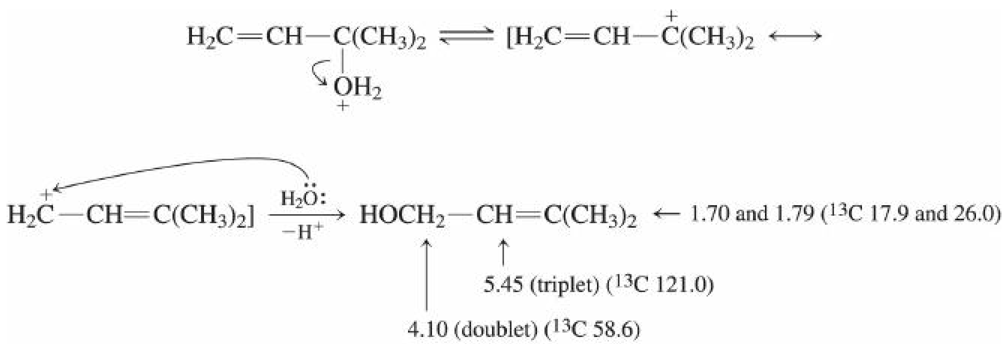
(f) π3→π4*$\pi_3 \rightarrow \pi_4{}^*$

71. These molecules contain only σσ and nn electrons, and their lowest unoccupied molecular orbitals are σ*.σ^*. The energy gaps between these orbitals are large, resulting in UV absorptions only at wavelengths shorter than 200 nm.200 nm.

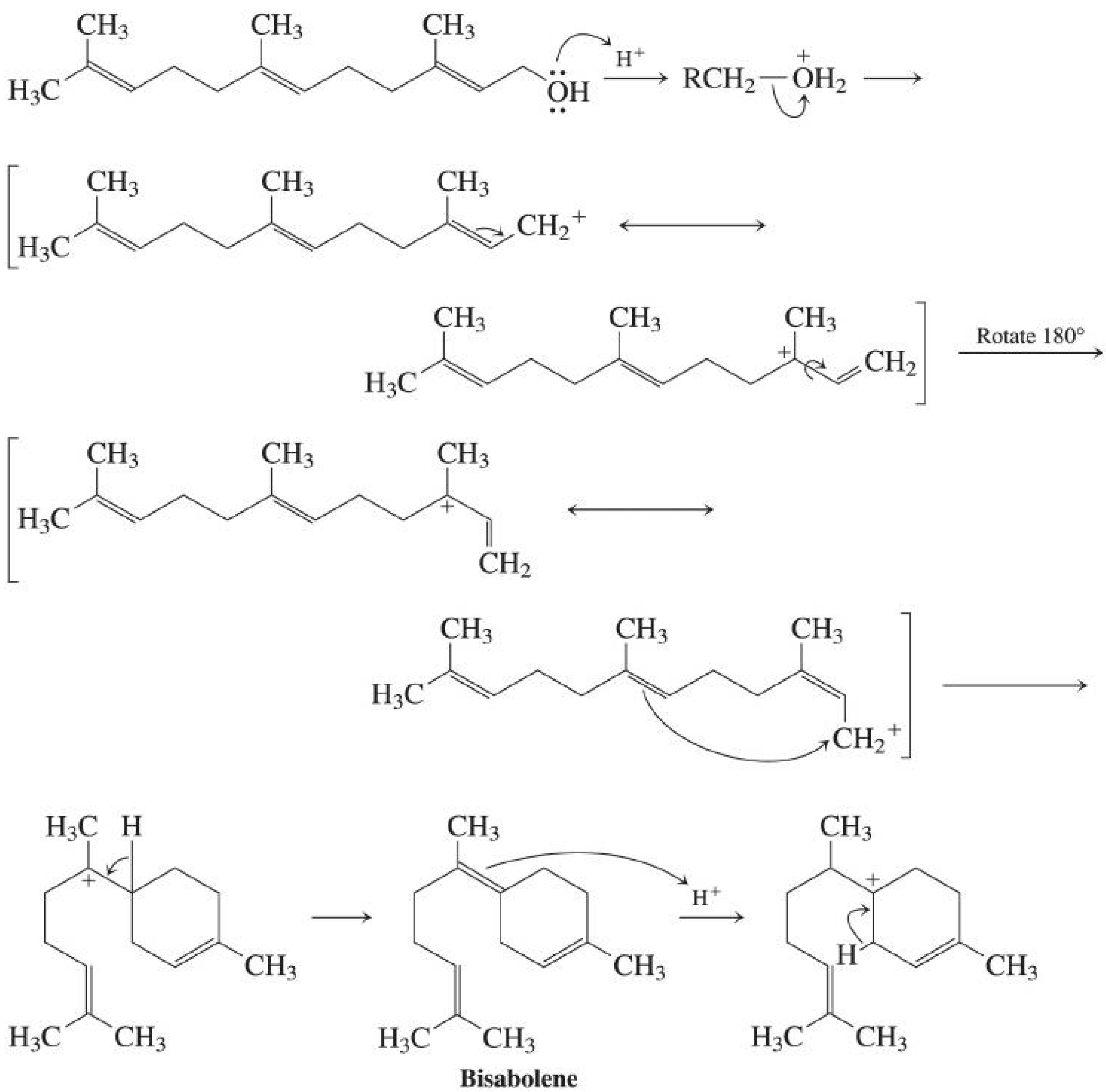
72. 1.95/(2×10—4)=9750; 0.008/(2×10—4)=40
$1.95/(2 \times 10^{-4})$â9750; $0.008/(2 \times 10^{-4}) = 40$

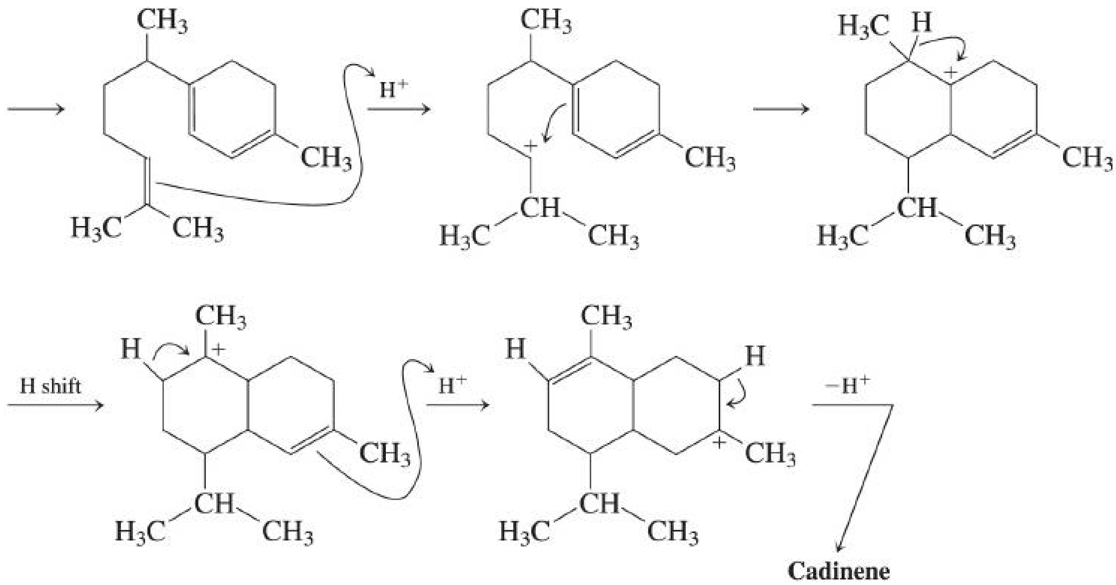
73. 

The new compound may be derived from ionization of the allylic alcohol upon extended contact with acid:

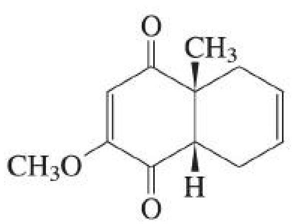


74.





75. The 1,2-addition product is kinetic, arising from attack of the nucleophile at the internal position of the intermediate allylic cation, where concentration of positive charge is highest. The 1,4-addition product contains an internal double bond, rendering it more stable.

76. 

This example illustrates several characteristic features of the cycloaddition process. Only the methyl-substituted double bond reacts. The methoxy-substituted one is too electron rich (see Exercise 14-14) to compete. The cis ring fusion retains the stereochemistry of the double bond in the dienophile. Finally, further cycloadditions of additional diene to the product are inhibited because neither of the double bonds in the latter is sufficiently electron deficient to react at an appreciable rate.

77. Ring closure of a 1,3-butadiene to a cyclobutene is the operative process. In order to bring the ends of the diene together to form the central bond in Dewar benzene, the mode of rotation must be disrotatory; otherwise, an impossibly strained product, with one cyclobutene ring substituted in a trans fashion by the other, would result. Disrotatory ring closure of a 1,3-diene is a photochemically promoted process (a photocyclization; see Figure 14-11). The resistance of this particular substituted Dewar benzene to reversion by ring-opening back to the corresponding benzene is in part a consequence of the reluctance of two tertiary alkyl groups to occupy adjacent (i.e., ortho) positions on a planar benzene ring. Dewar benzenes are severely bent structures (make a model!), and the adjacent *tert*-butyl groups in this one are well out of each other's way as a result.

78.

(c)

79.

(b)

80.

(b)

81.

(b)

15 Benzene and Aromaticity: Electrophilic Aromatic Substitution

The most obvious feature of benzene is its superficial similarity to a molecule containing ordinary double bonds. Benzene and its derivatives are special, however, as a result of the new concept of aromatic stabilization covered in the early sections of this chapter. The electronic structure of benzene, which gives this compound unusual stability relative to that of an ordinary triene, profoundly affects benzene's chemistry. Each time you encounter a new property or chemical reaction of a benzene derivative, ask yourself, "How would the properties or reactions of a simple alkene (or diene or triene) compare with this?" See if the differences in behavior of benzene and alkenes make sense to you on thermodynamic grounds. After you've done that, you will be in a better position to learn the material in this and the next chapter more thoroughly, with a more balanced overview of the entire topic.

Benzene and its derivatives make up one of the most important classes of organic compounds (after carbonyl compounds and alcohols). There is, accordingly, a lot of relatively significant material presented in these chapters concerning them.

Other kinds of aromatic compounds exist besides simple benzene derivatives. Two common types will be covered in this chapter: polycyclic fused benzenoid hydrocarbons and other cyclic conjugated polyenes with either more or less than six carbons in the ring. In Chapter 25, a third common class, heterocyclic aromatic compounds, will be presented.

Outline of the Chapter

15-1 Nomenclature

A few new twists.

15-2, 15-3 Structure of Benzene: A First Look at Aromaticity

What makes it special.

15-4 Spectral Properties

15-5 Polycyclic Aromatic Hydrocarbons

15-6, 15-7 Other Cyclic Polyenes: Hückel's Rule

Getting to the heart of the matter: Whence aromaticity?

15-8, 15-9, 15-10 Electrophilic Aromatic Substitution

The basic "core" reactions in benzene chemistry.

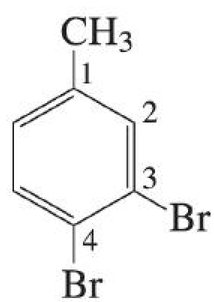
15-11, 15-12, 15-13 Friedel-Crafts Reactions

Making carbon-carbon bonds to benzene rings.

Keys to the Chapter

15-1. Nomenclature

The systematic (IUPAC) naming system for benzenes coexists with an assortment of names that have been in common use for over a century and show no signs of going away. So, whether anyone likes it or not, terms like aniline (for benzenamine) and styrene (for ethenylbenzene) are, and will probably always be, the ones people use, both in speaking as well as writing about these compounds. Indeed, both *Chemical Abstracts* and the *IUPAC* have given up on benzenamine altogether and fully accepted aniline as the name of the compound. In addition to these common names for simple substituted benzenes, a special system exists exclusively for benzenes with **exactly two** substituents on the benzene ring: the "ortho/meta/para" system. Note, carefully, that this method is **never** to be used for benzenes with more than two ring substituents: For those, a name with proper numbering is required. (It is okay, however, to use numbering instead of ortho/meta/para for a benzene with two substituents.) Finally, when numbering around the benzene ring, be careful where you start: If you use one of the special names for a monosubstituted benzene as your parent (e.g., phenol, aniline, benzaldehyde, toluene), C1C1 is **always** the carbon containing the substituent implied by the parent (even if starting somewhere else would result in smaller numbers). For example, this compound is 3,4-dibromotoluene (1,2-dibromotoluene would be nonsense, although 1,2-dibromo-4-methylbenzene is the correct **IUPAC** name).



15-2 and 15-3. Structure of Benzene: A First Look at Aromaticity; Molecular Orbitals

Aromaticity as a special property of molecules like benzene is covered in these sections. Structural, thermodynamic, and electronic considerations are presented and serve as an introduction to the more general discussion that is presented in Section 15-6. For now, note simply that the aromaticity of benzene is reflected in (1) its symmetrical structure (as a resonance hybrid), (2) its unexpectedly enhanced thermodynamic stability, and (3) its unusual electronic structure with a completely filled set of strongly stabilized bonding molecular orbitals.

15-4. Spectral Properties

The spectroscopy of benzene is a logical consequence of its structural and electronic properties. Special features in the spectra make the identification of benzenes relatively easy. As has been the case with other compounds, NMRNMR is most useful, followed by IRIR and UVUV spectroscopy. A special feature in the infrared spectrum of a benzene derivative is the C—H$\mathrm{C—H}$ out-of-plane bending vibration pattern, which provides information concerning the arrangement of substituents around the ring. This information is not always as easily derived from the NMRNMR spectrum. Ultraviolet spectroscopy can be suggestive but not definitive regarding the presence of a benzene derivative. If the presence of the benzene ring is already known from, e.g., NMR,$\mathrm{NMR},$ the UVUV spectrum can be useful in deciding whether it is conjugated with one or more ππ bond-containing groups.

15-5. Polycyclic Aromatic Hydrocarbons

The fusion of benzene rings leads to a large class of polycyclic hydrocarbons, beginning with naphthalene as the simplest (and only bicyclic) example.

15-6 and 15-7. Other Cyclic Polyenes: Hückel’s Rule

Really very simple. For a molecule to be aromatic, it must have (4n + 2) π $(4n + 2)\pi$ electrons contained in a complete, unbroken circle of pp orbitals.

15-8, 15-9, and 15-10. Electrophilic Aromatic Substitution: Halogenation, Nitration, and Sulfonation of Benzene

These sections introduce the main type of chemistry displayed by benzene rings. The keys to this material are in Section 15-8. You are given a general mechanism (which is repeated later in specific form for each type of reaction that is presented). More important for understanding the concepts, you are given information concerning the energetics of this reaction. Pay close attention to Figure 15-20, Exercise 15-22, and the data in Section 15-8. After you understand this basic material, you will find the specific reactions a little easier to learn. Remember that benzene is stabilized by its special aromatic form of resonance. It is **less reactive** toward electrophiles than are ordinary alkenes; and therefore only **strong electrophiles,** sometimes generated in strange ways, will attack the $\pi\pi$ electrons in benzene rings. These electrophiles, and the ways they are generated, have to be memorized so that you can use these reactions later on in synthesis problems.

15-11, 15-12, and 15-13. Friedel-Crafts Reaction

Because carbon – carbon bond formation is such an important part of organic synthesis, the attachment of carbon electrophiles to benzene rings is of special significance among the reactions in this chapter. Ordinary carbocations and carbocationlike species are the simplest types of carbon electrophiles, but their use in Friedel-Crafts alkylation has limitations. Rearrangements of the carbon electrophile often occur, and it is hard to prevent multiple alkylation from happening to a single benzene ring. Friedel-Crafts *acylation,* via the acylium ion [R—C+═O¨:↔R—C≡O+:],

$[R\text{—}\overset{+}{C}\text{â}\,\ddot{O}\!: \leftrightarrow R\text{—}C{\equiv}\overset{+}{O}\!:]$, is not subject to these drawbacks and, whenever possible, is the preferred method for attachment of a carbon unit to a benzene ring.